

SafeMine — Landmine Detection Model

YOLOv8m Training Report

Training Configuration

Model:	yolov8m.pt
Epochs:	75
Image size:	640 px
Batch size:	8
Training time:	15h 16m 42s
Random seed:	0

Hardware

GPU:	NVIDIA GeForce RTX 4070 Laptop GPU
VRAM:	8 GB
PyTorch:	2.6.0+cu124
Ultralytics:	8.4.117
CUDA:	12.4

Dataset

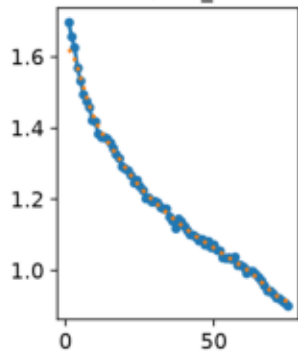
Path:	landmine_combined (C:/Users/Delta/Downloads/...)
Train images:	5,723
Val images:	455
Test images:	221
Classes:	1 (Landmine)
Image size:	640x640 RGB

Final Metrics

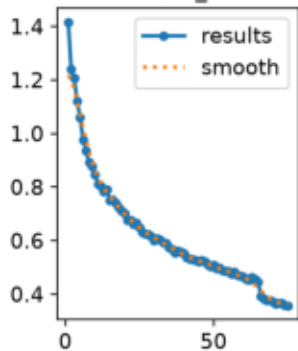
Metric	Validation	Test
Precision	0.9414	0.9655
Recall	0.9537	0.9572
mAP@50	0.9583	0.9718
mAP@50-95	0.6239	0.6360

Note: a missed mine is a safety failure, a false positive is an annoyance.

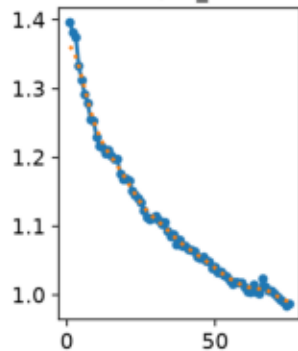
train/box_loss



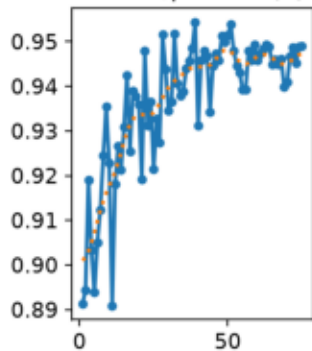
train/cls_loss



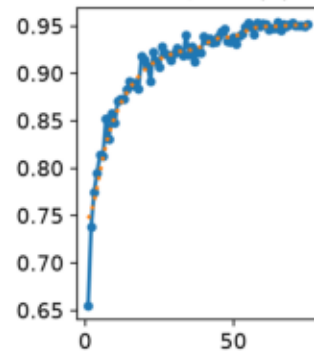
train/df_l_loss



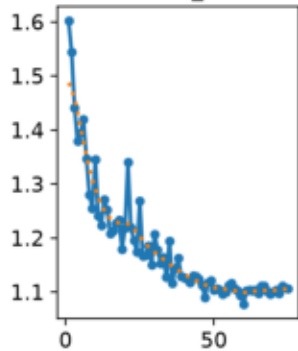
metrics/precision(B)



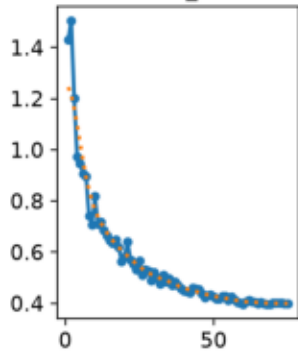
metrics/recall(B)



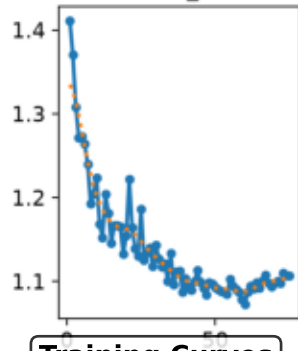
val/box_loss



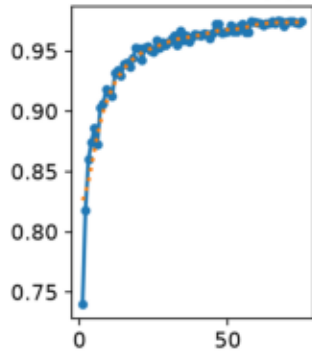
val/cls_loss



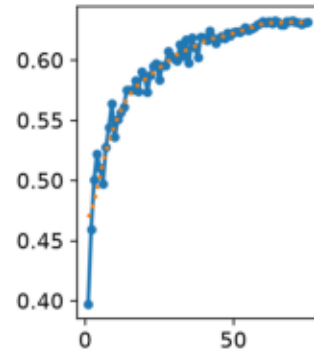
val/df_l_loss



metrics/mAP50(B)

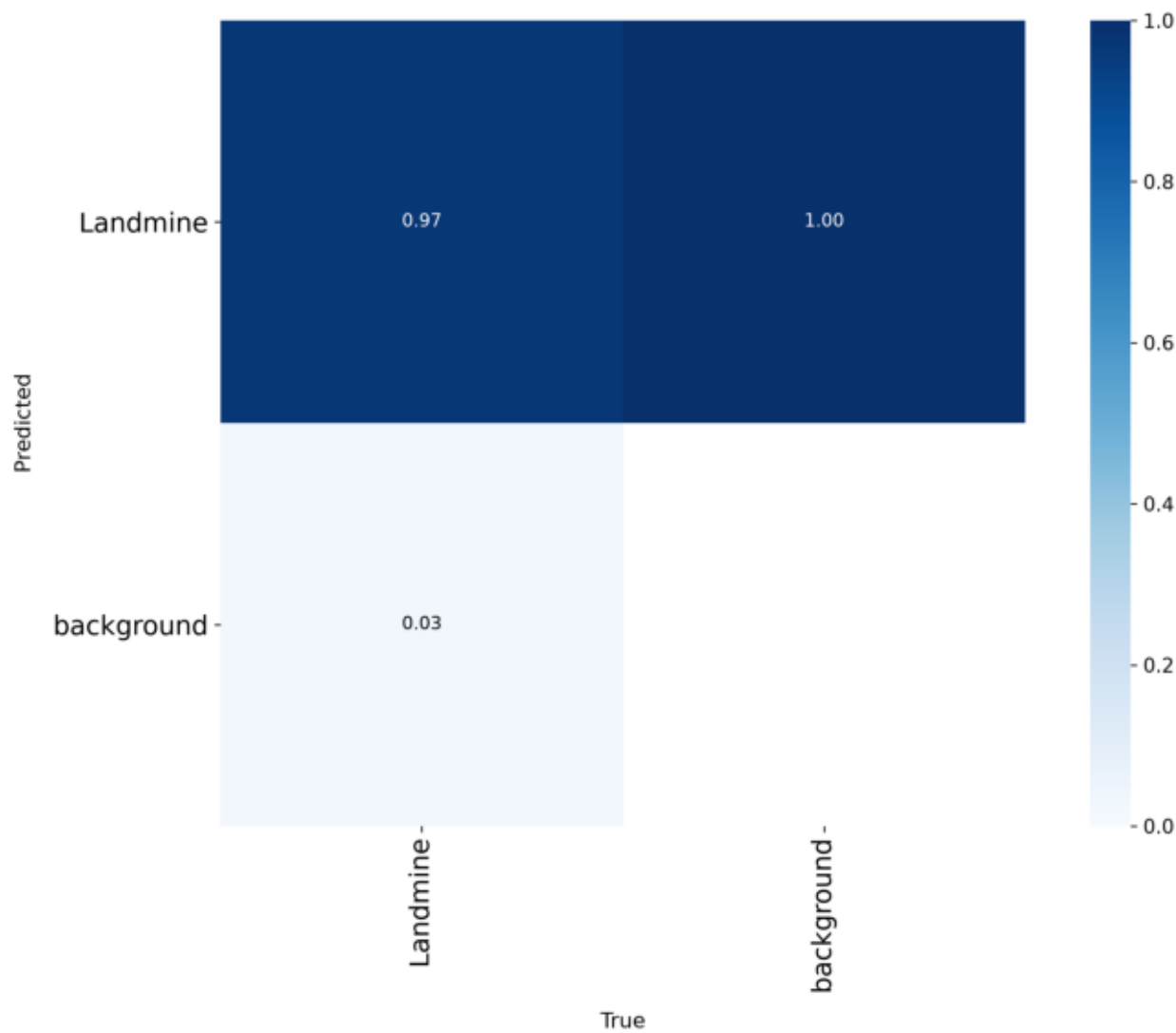


metrics/mAP50-95(B)



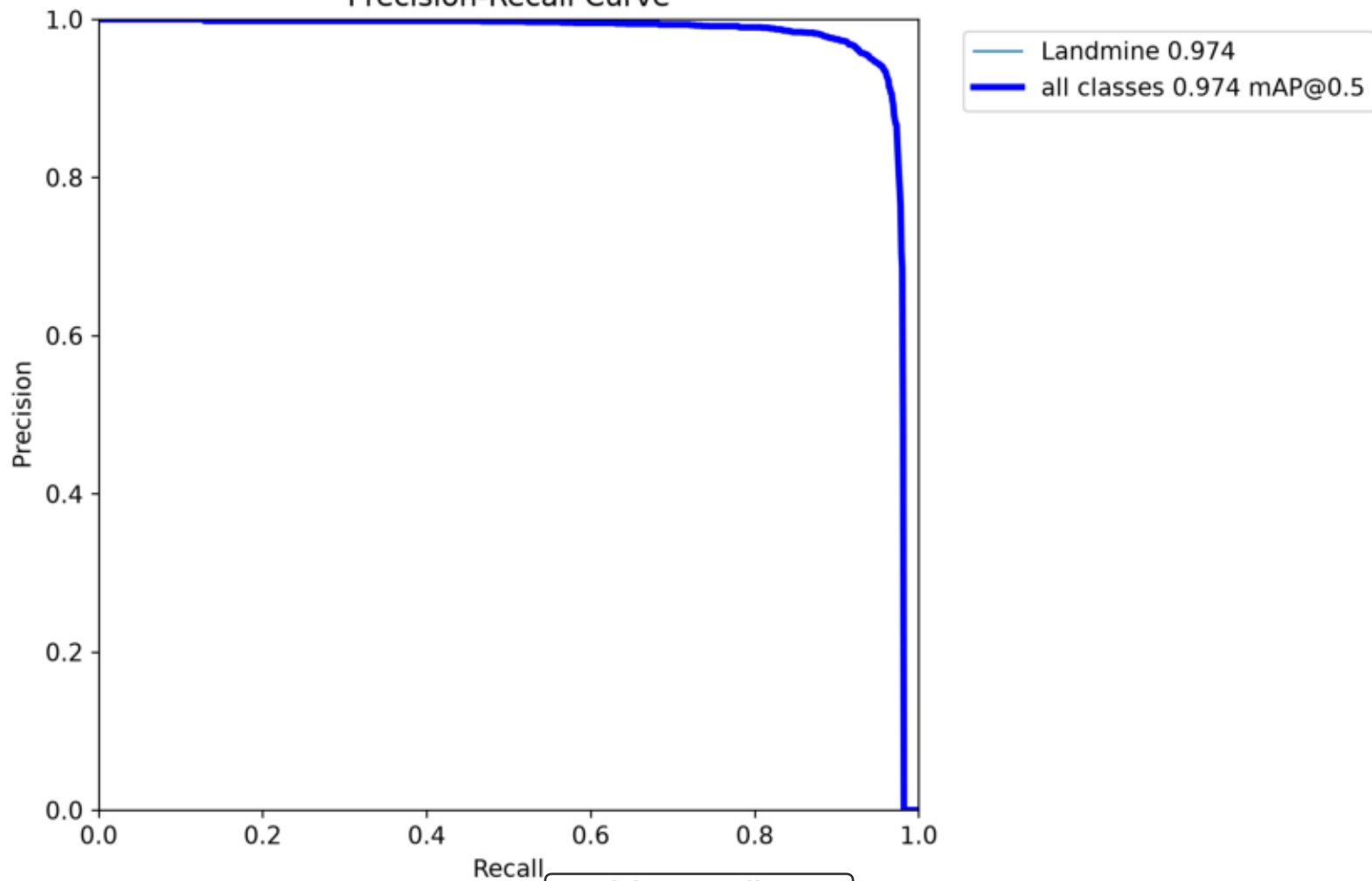
Training Curves

Confusion Matrix Normalized



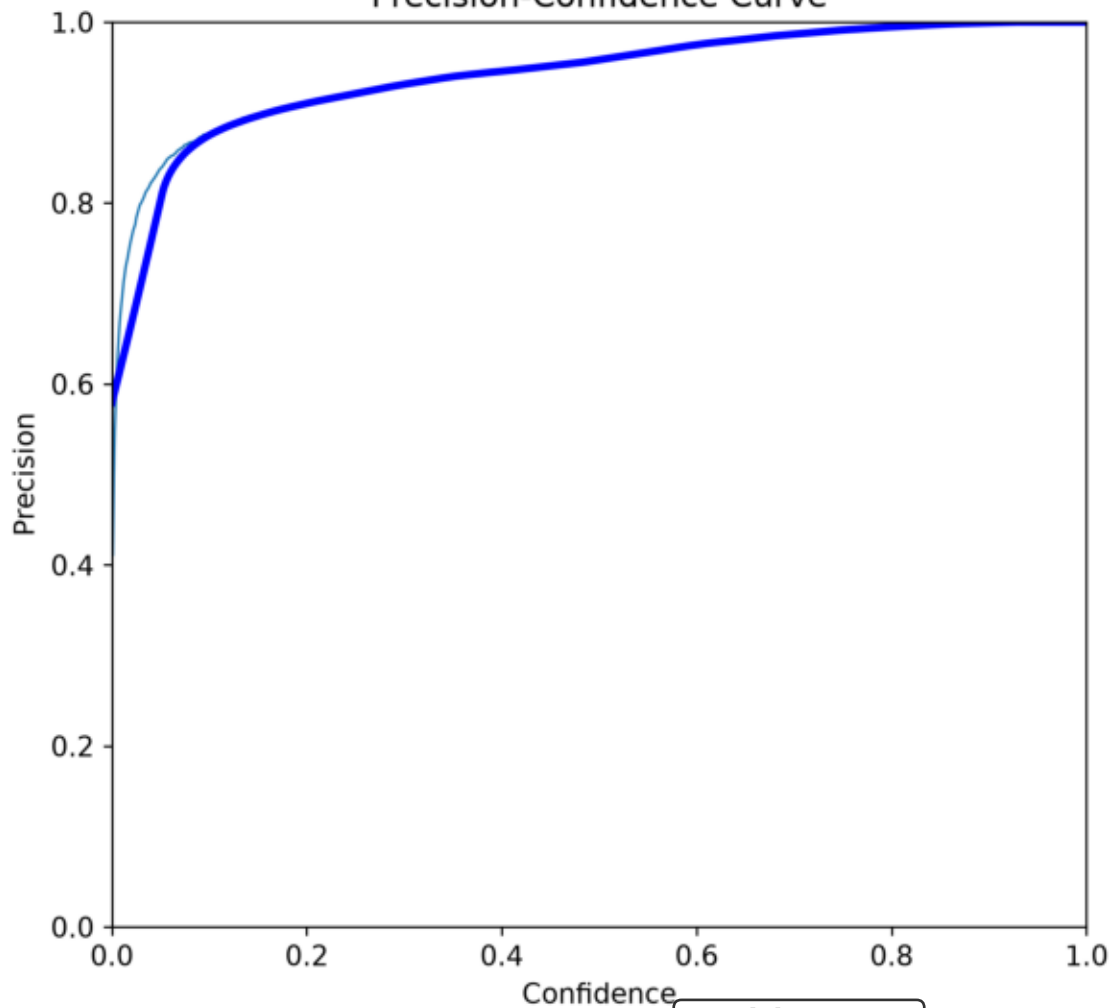
Confusion Matrix (Normalized)

Precision-Recall Curve



Precision-Recall Curve

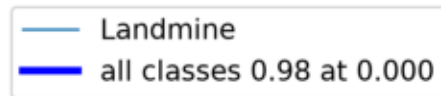
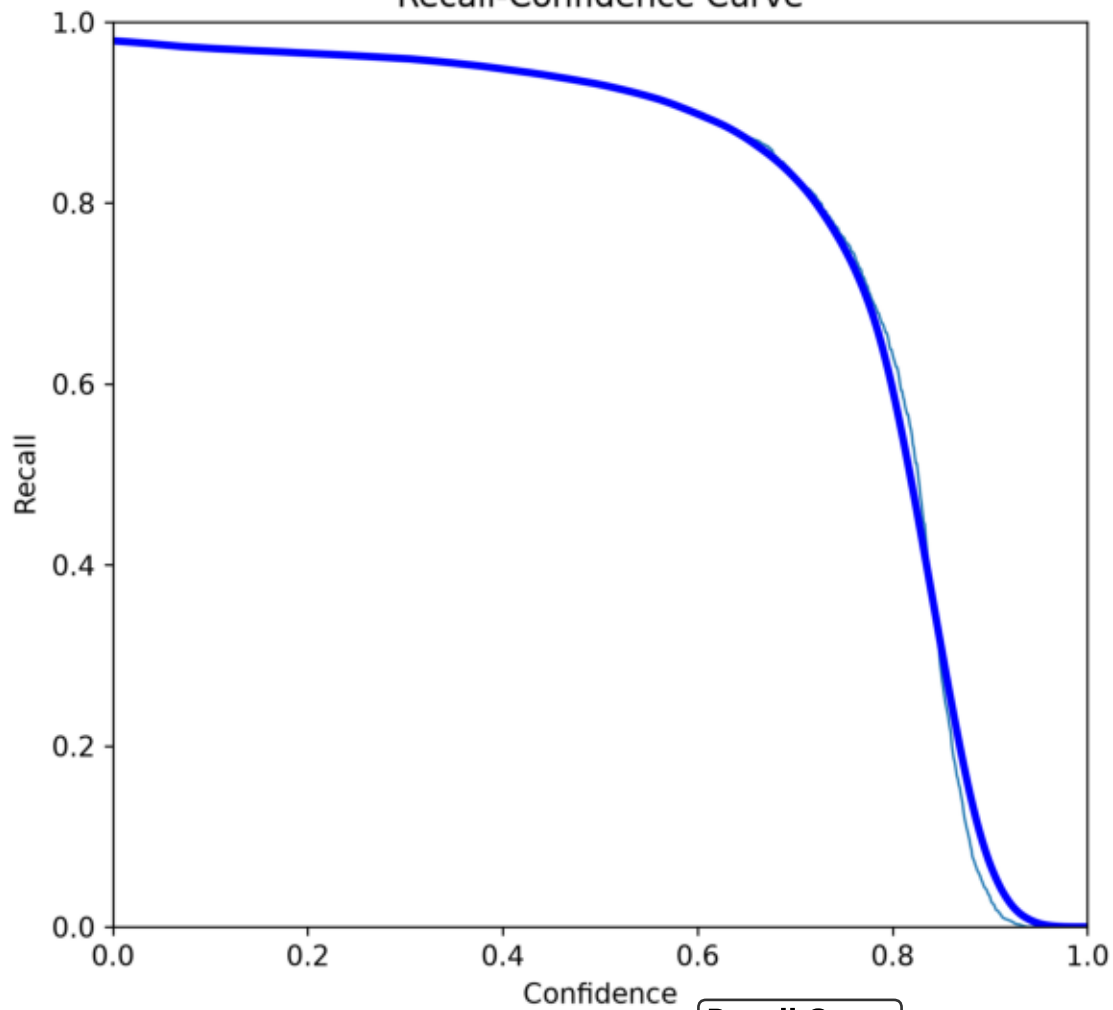
Precision-Confidence Curve



Landmine
all classes 1.00 at 0.923

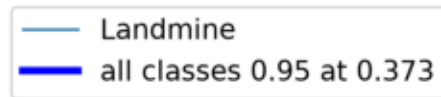
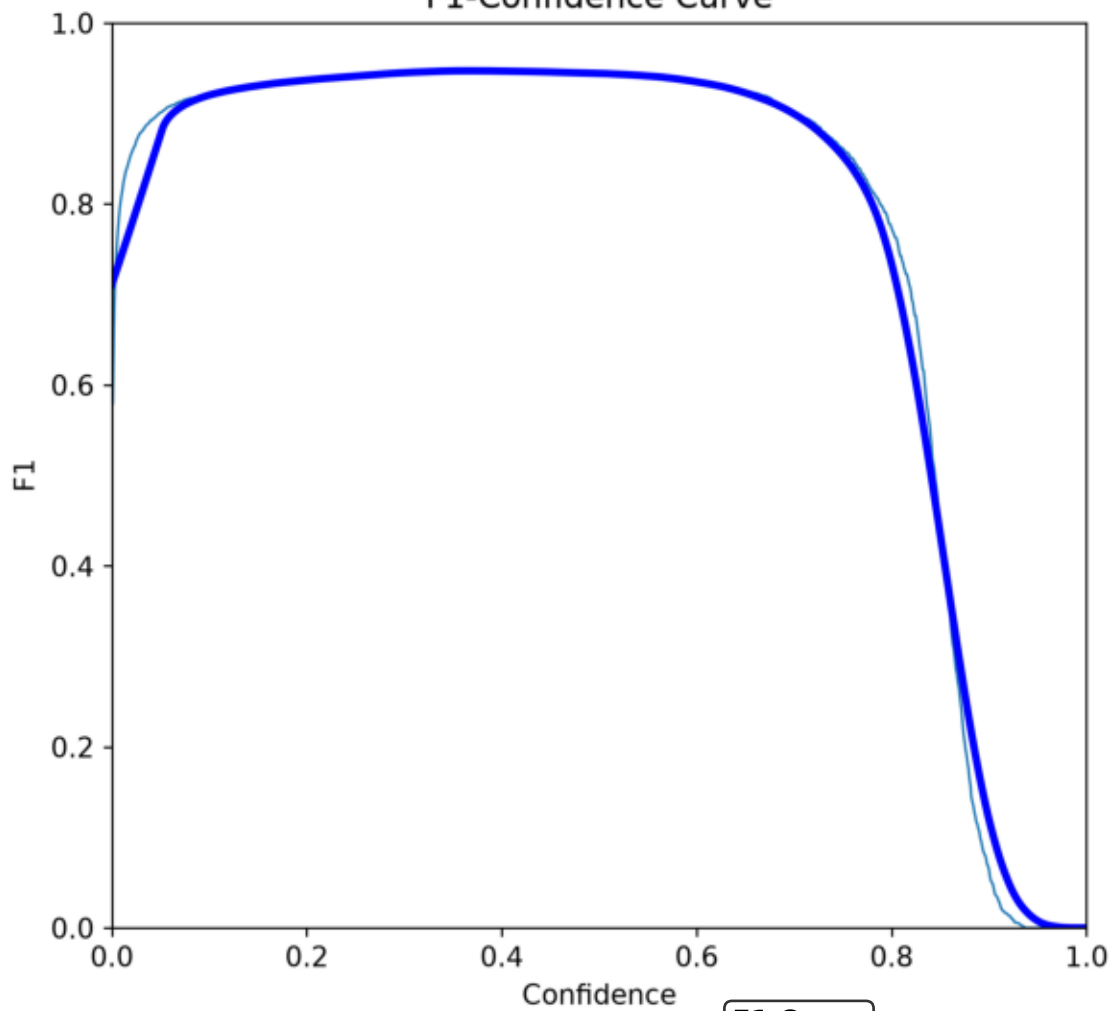
Precision Curve

Recall-Confidence Curve

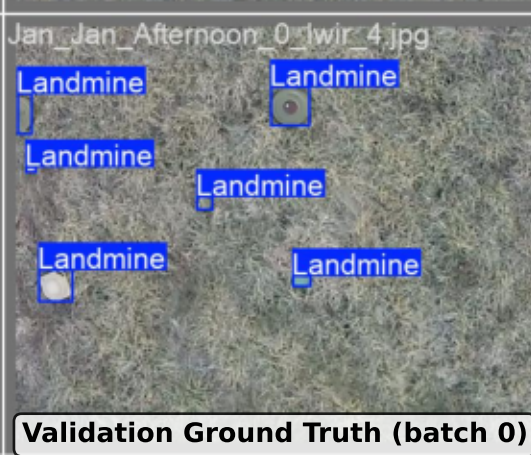
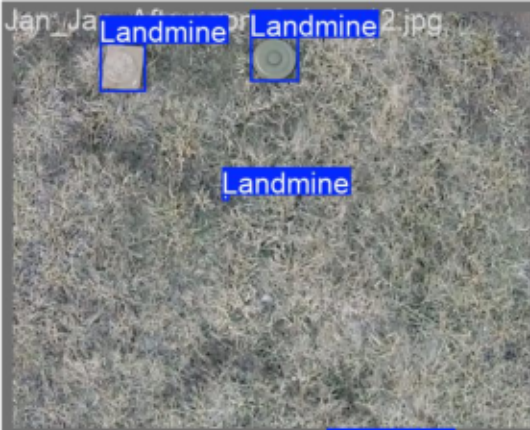


Recall Curve

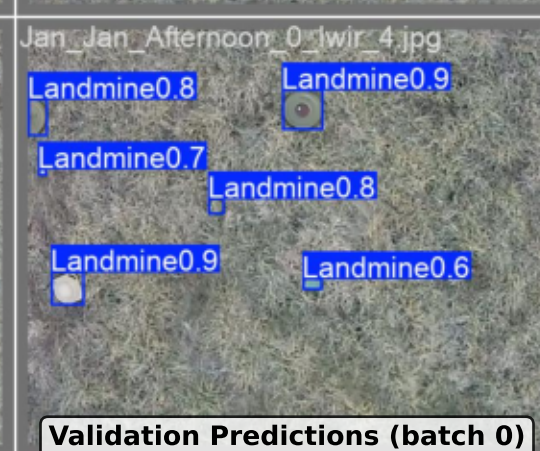
F1-Confidence Curve



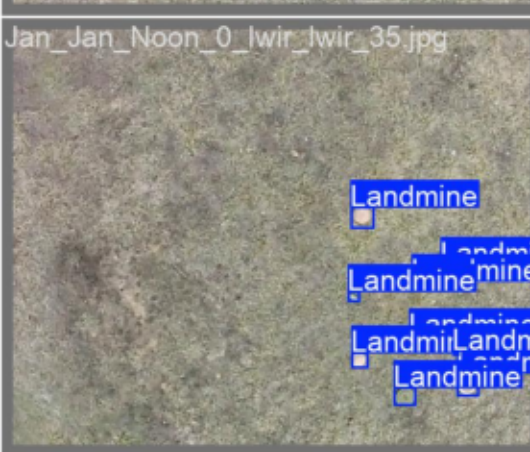
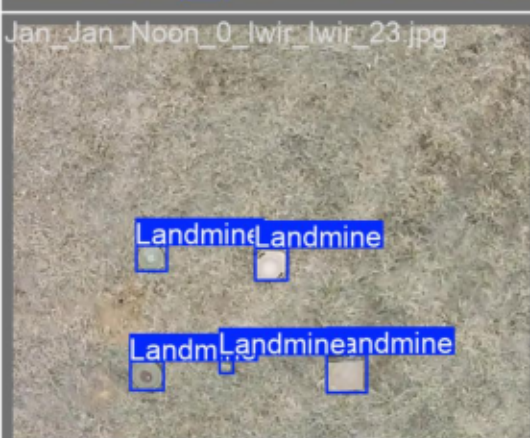
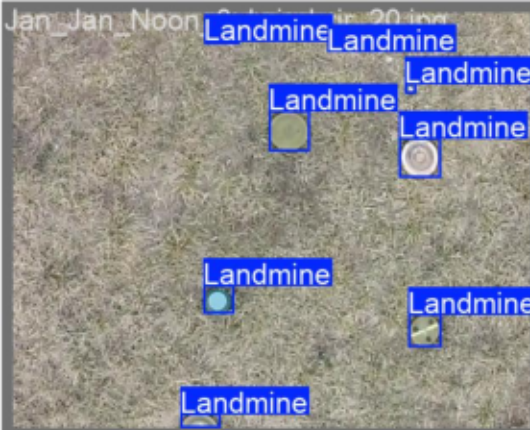
F1 Curve



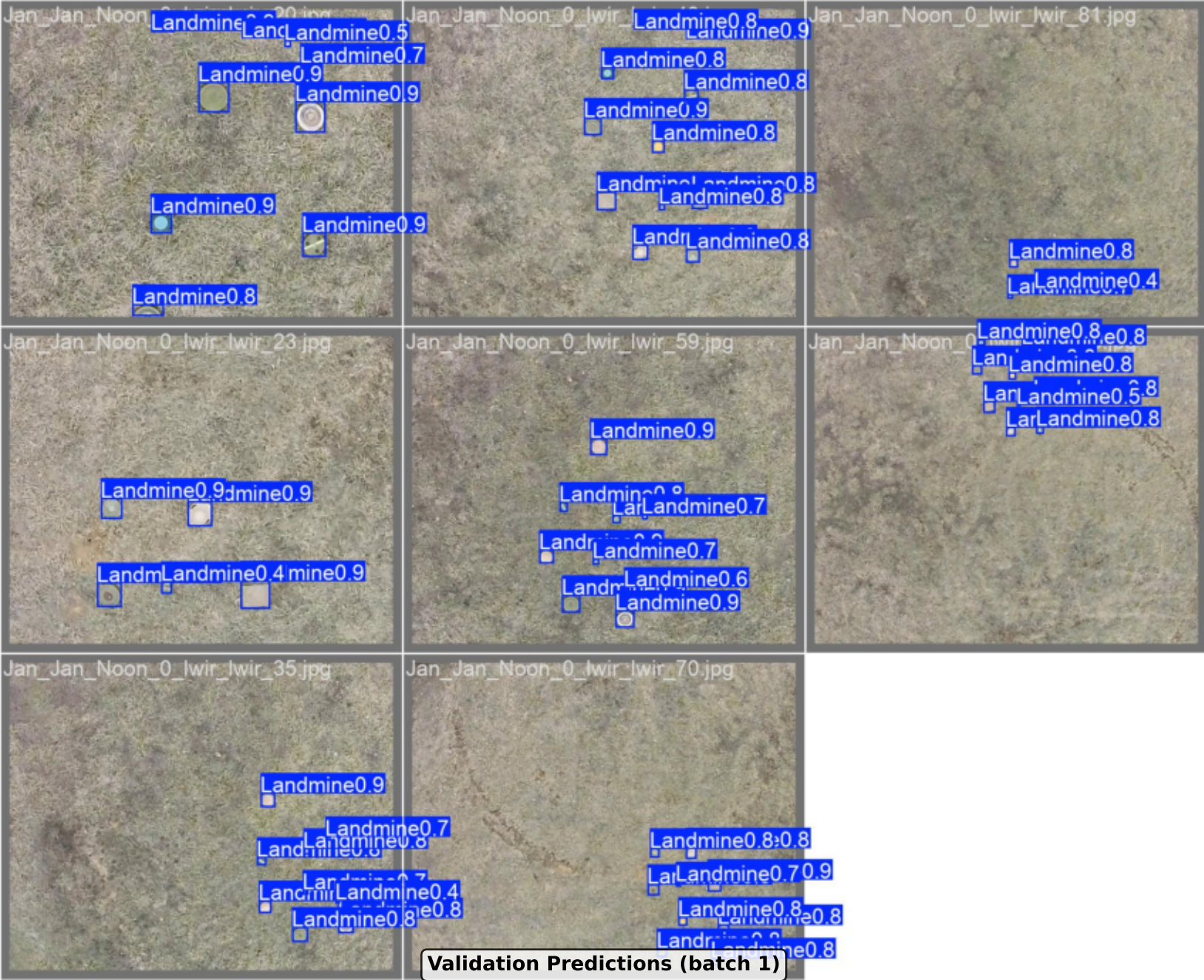
Validation Ground Truth (batch 0)

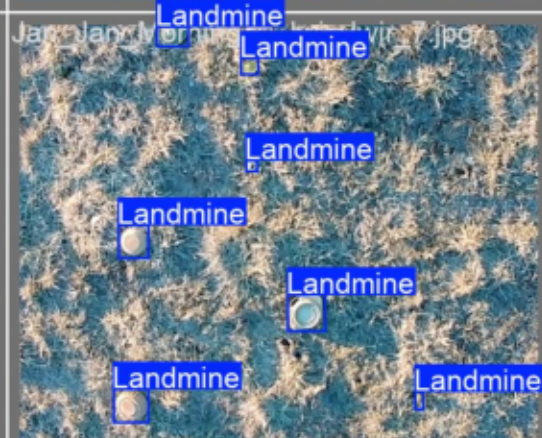
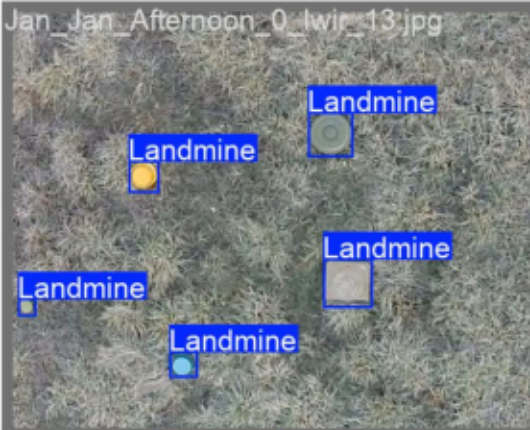


Validation Predictions (batch 0)



Validation Ground Truth (batch 1)



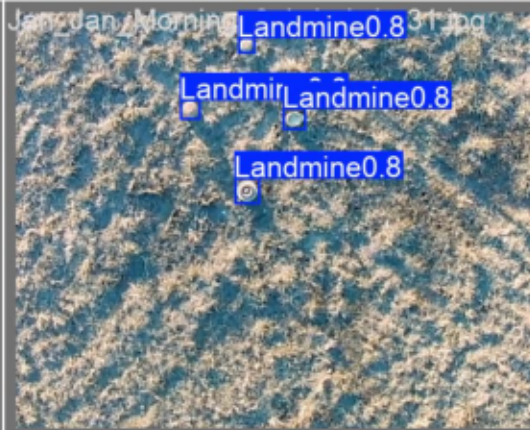


Test Ground Truth (batch 0)

Jan_Jan_Afternoon_0_lwir_13.jpg



Jamdan Morning Landmine0.831.jpg



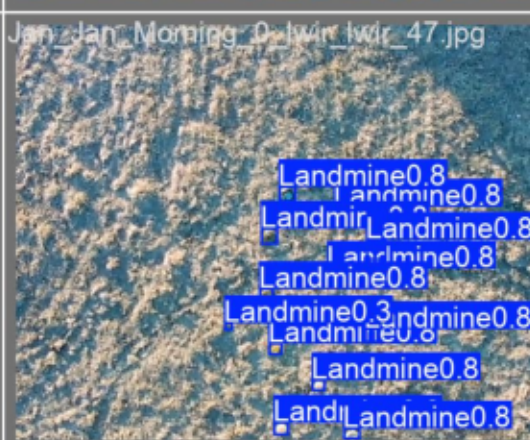
Jan 14 11:01 AM Landmine08081401.jpg 58 KB



Landmine0.8



Jan-Jan Morning Delwir Iwdr_47.jpg



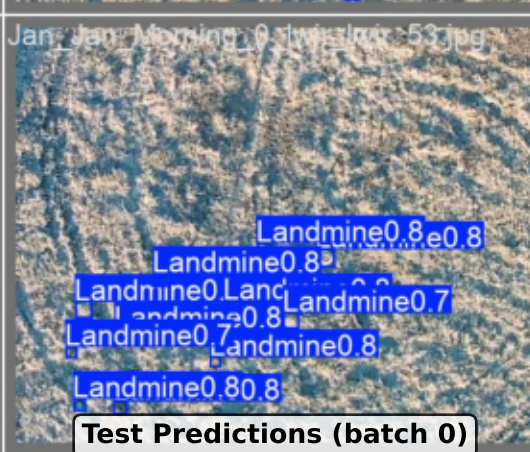
Landmine0.8



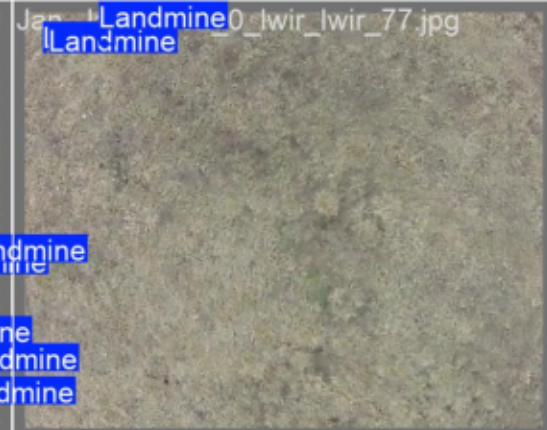
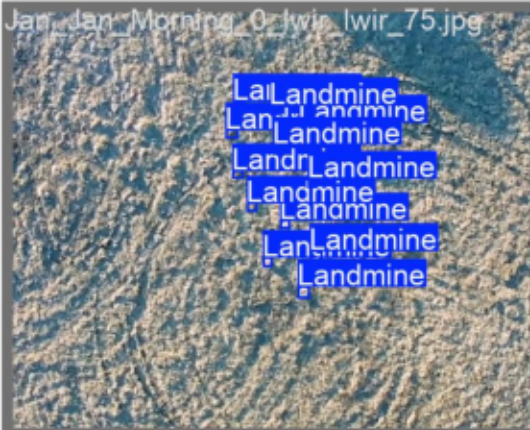
Jan_Jan_Afternoon_Landmine0.4_01wlr_55.jpg



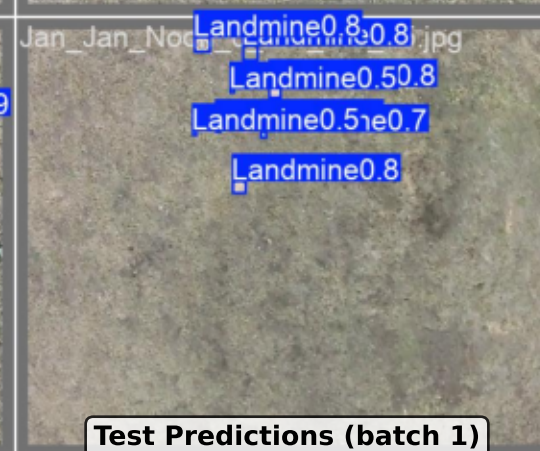
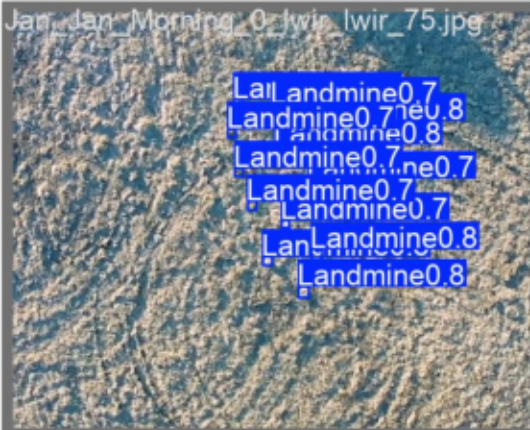
Jan Jan Morning 0 twin lock 53 jug



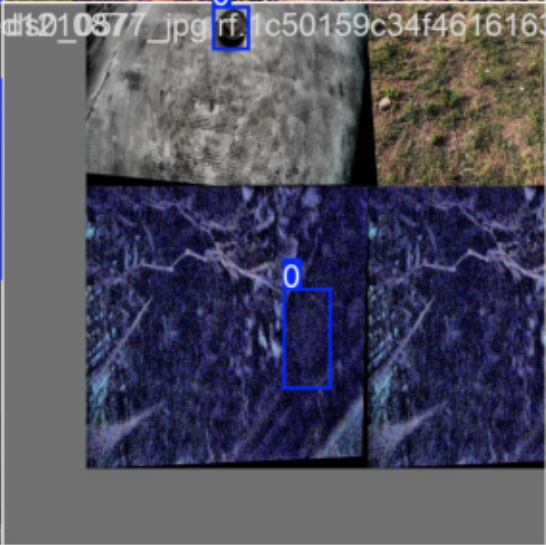
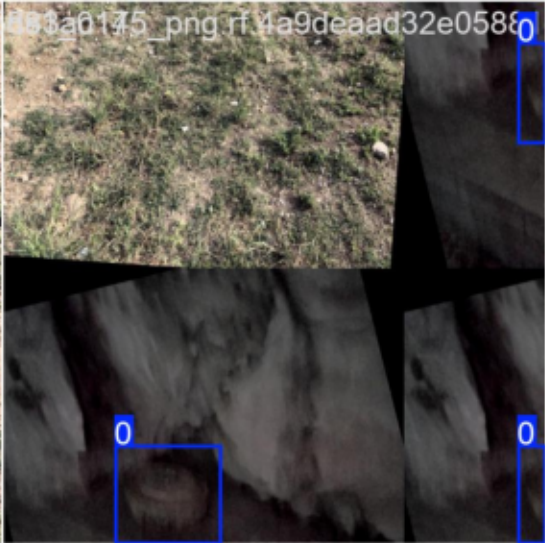
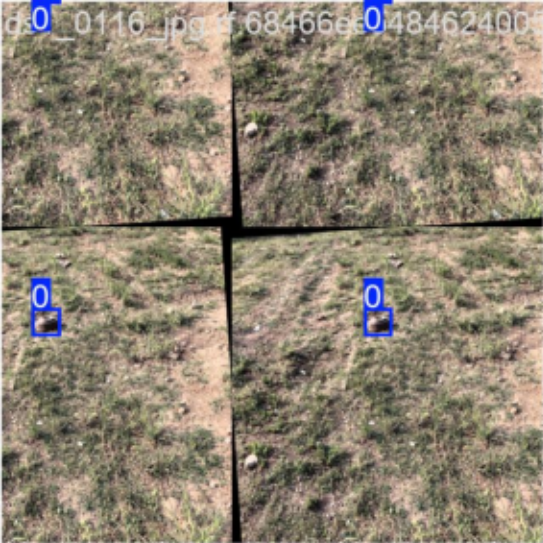
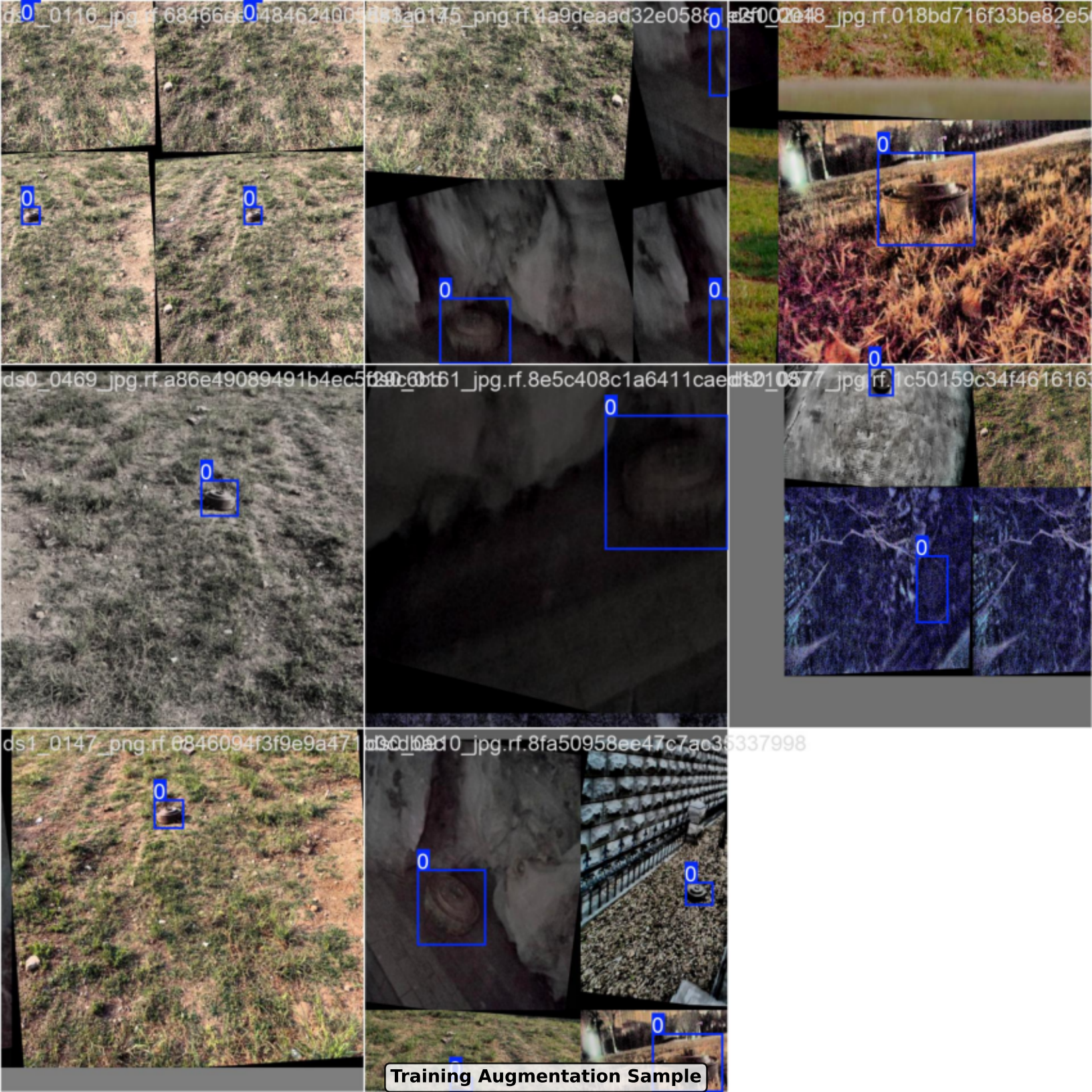
Test Predictions (batch 0)



Test Ground Truth (batch 1)



Test Predictions (batch 1)



Training Augmentation Sample